

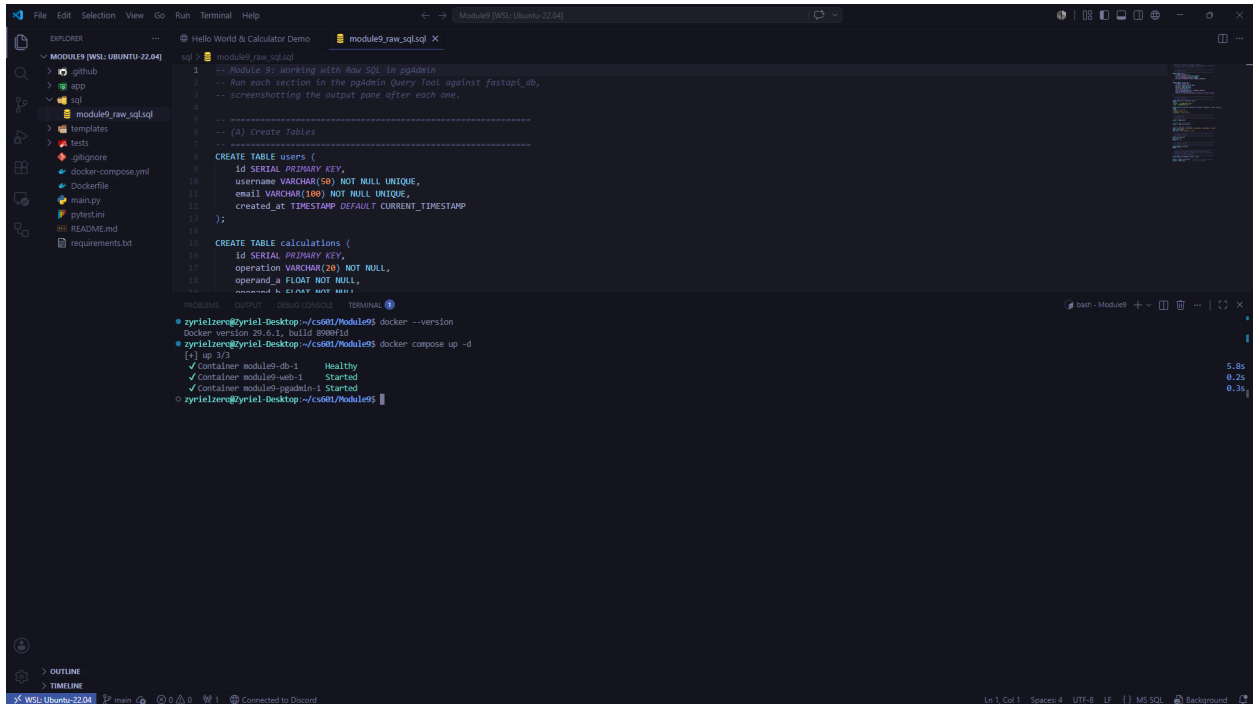


Figure 1. docker compose up -d bringing up all three services: the db container reports Healthy and web and pgadmin report Started. The project structure with docker-compose.yml, Dockerfile, and sql/module9_raw_sql.sql is visible in the explorer.

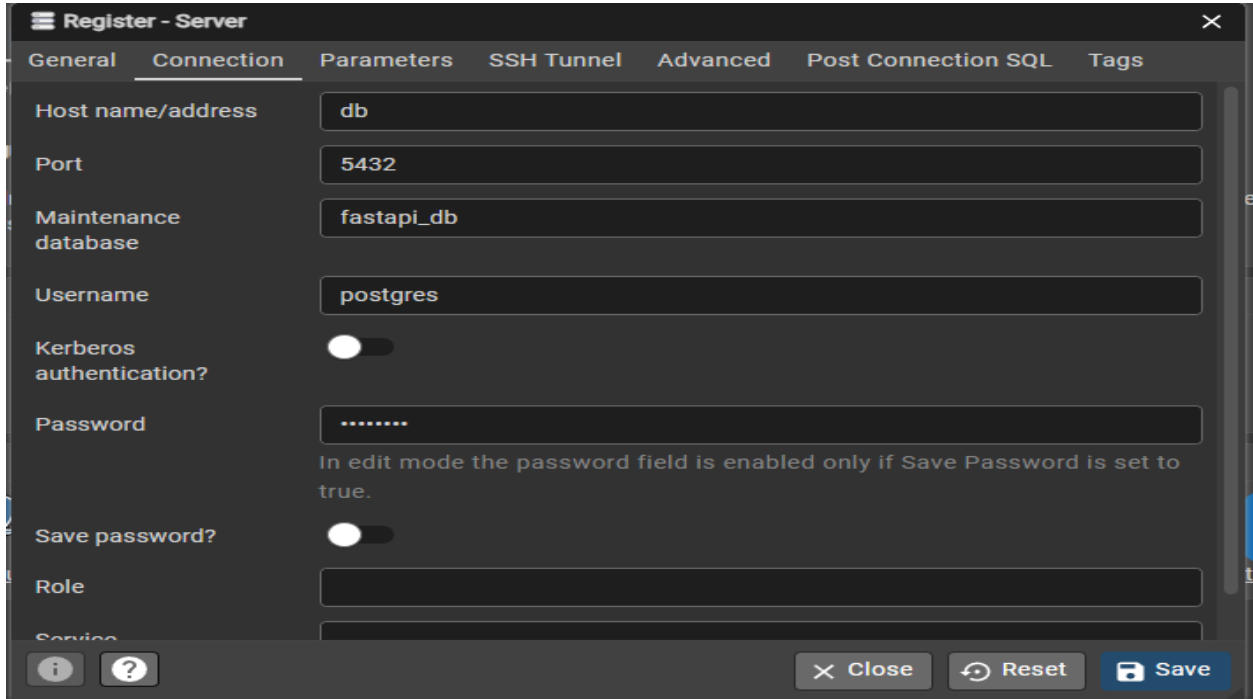


Figure 2. Registering the PostgreSQL server in pgAdmin. Host is the Compose service name db, port 5432, maintenance database fastapi_db, user postgres.

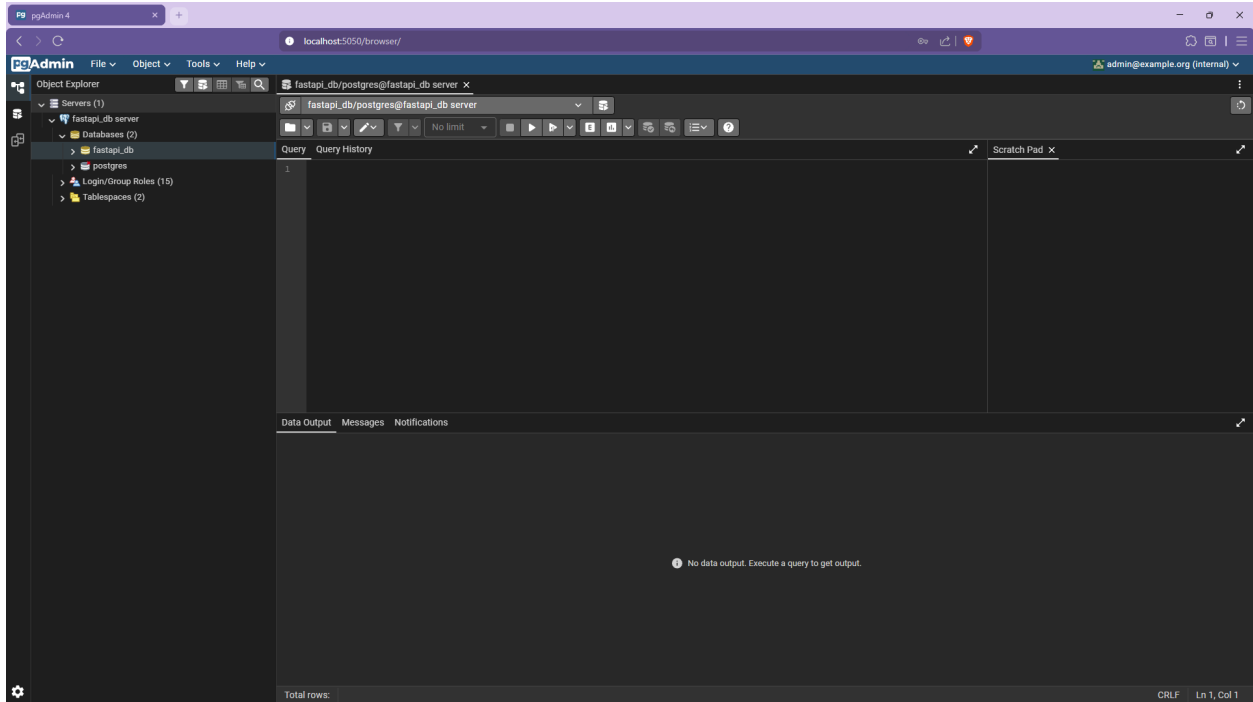


Figure 3. Query Tool connected to fastapi_db, ready to execute SQL.

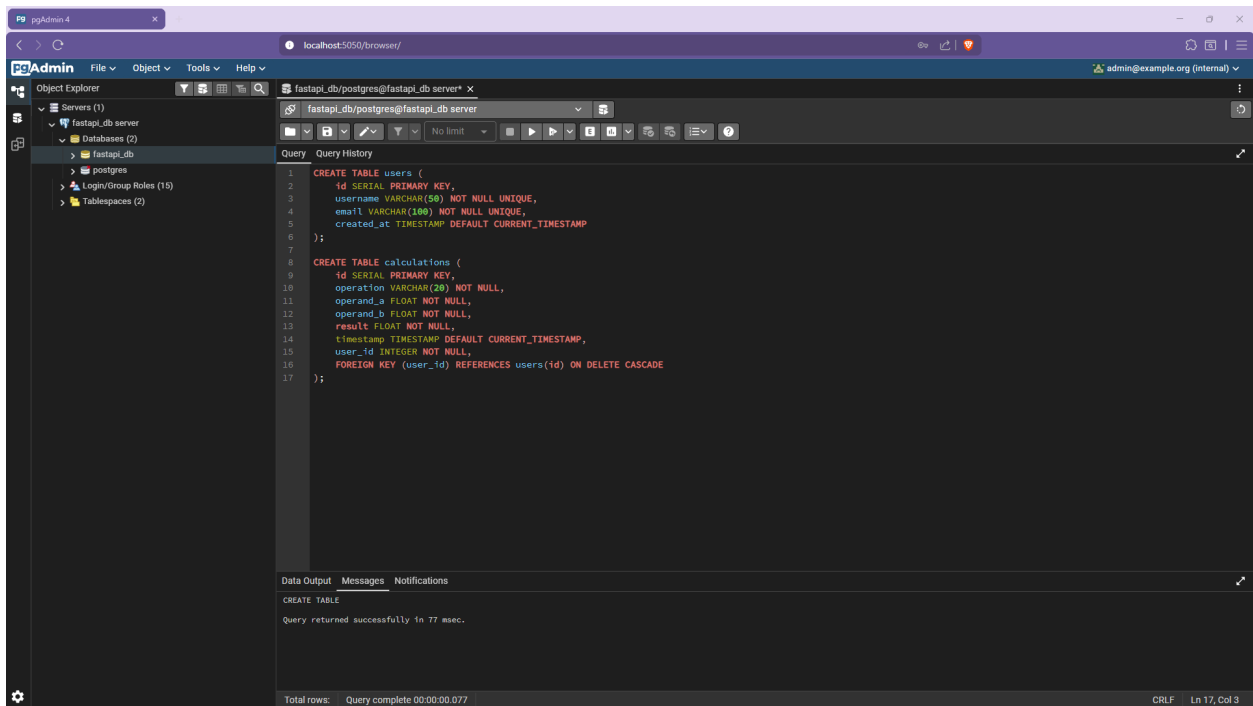


Figure 4. Both CREATE TABLE statements executed. Messages tab confirms: CREATE TABLE, query returned successfully.

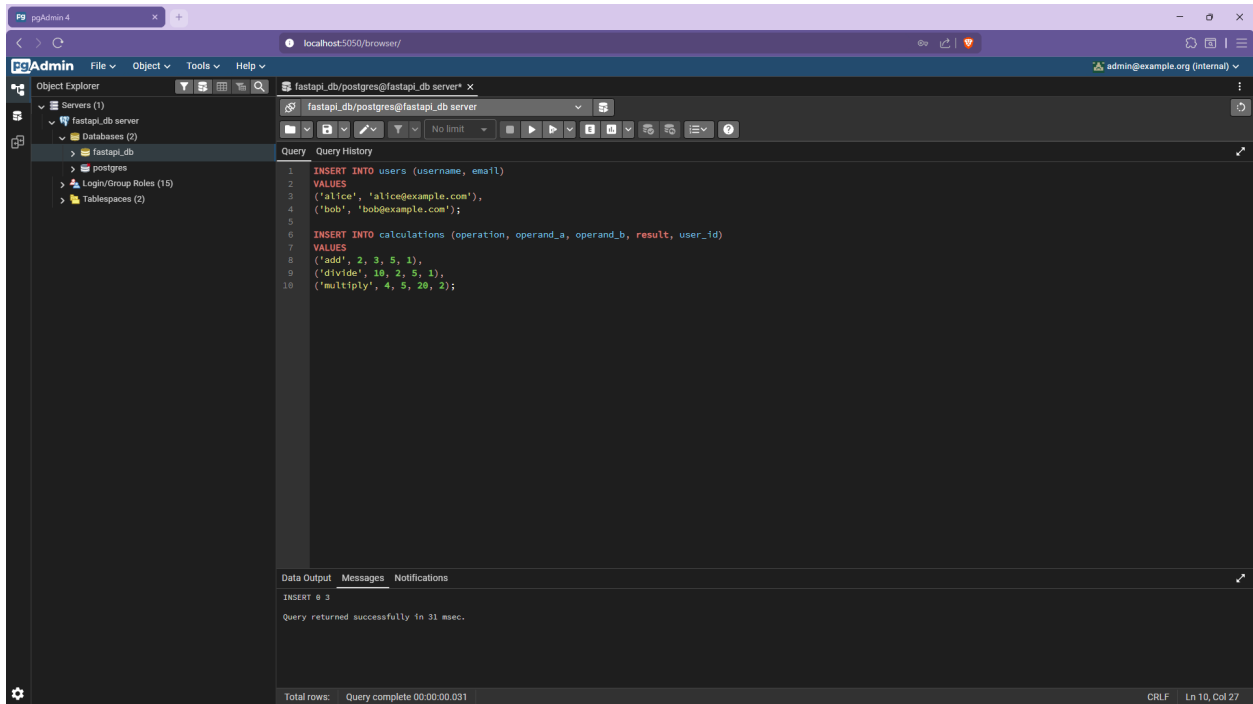


Figure 5. INSERT statements executed. Messages tab confirms INSERT 0 3 for the calculations batch.

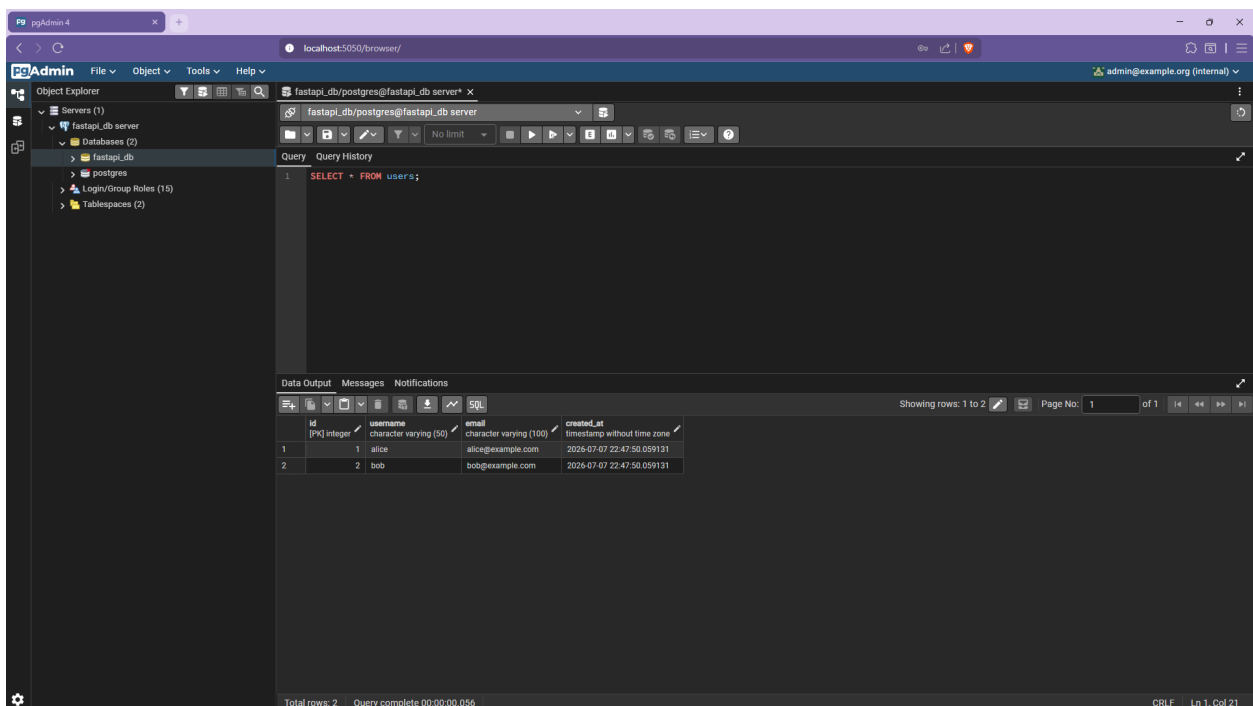


Figure 6. SELECT * FROM users returns both rows: alice and bob.

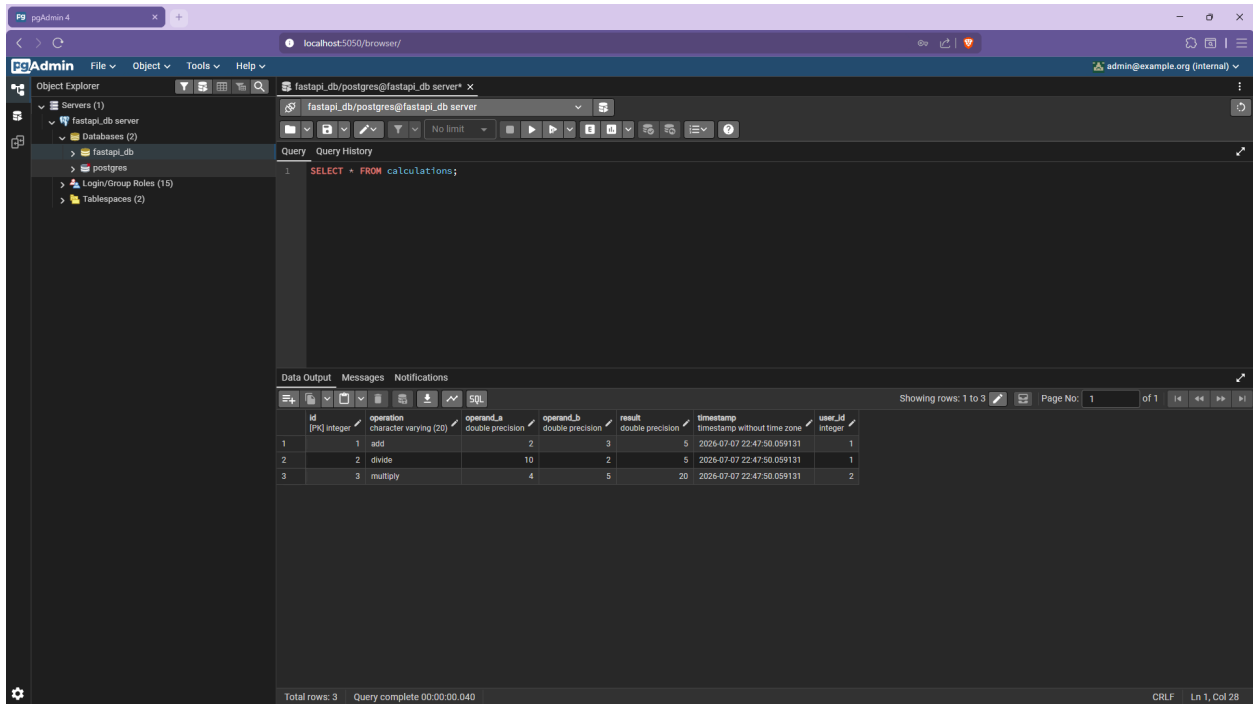


Figure 7. `SELECT * FROM calculations` returns all three rows with their `user_id` foreign keys.

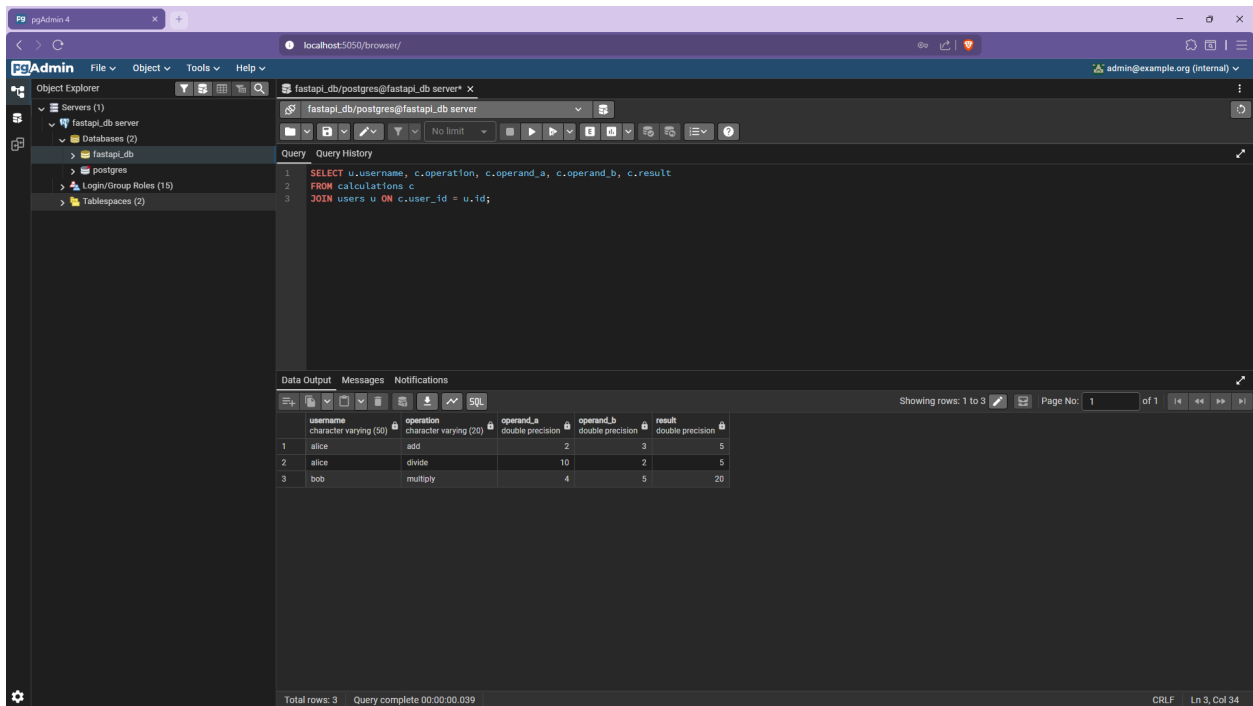


Figure 8. Join of calculations to users. Alice owns the add and divide rows; bob owns multiply. This demonstrates the one-to-many relationship.

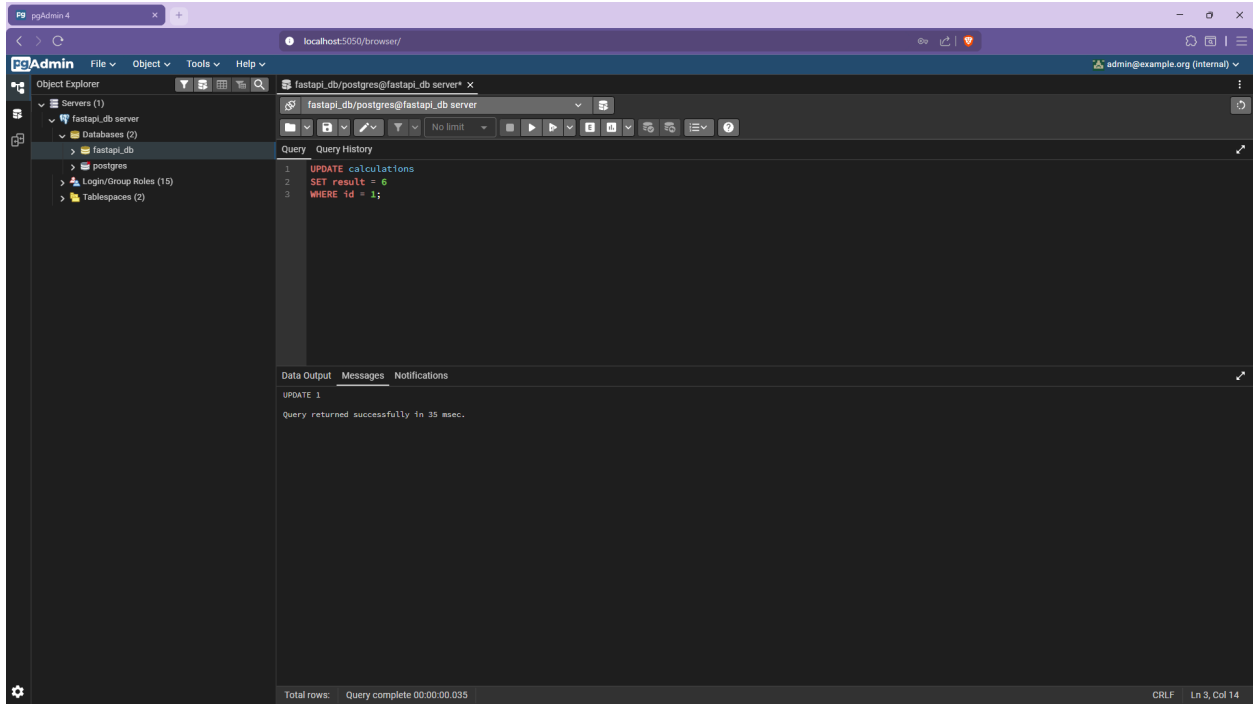


Figure 9. UPDATE executed. Messages tab confirms UPDATE 1, exactly one row affected.

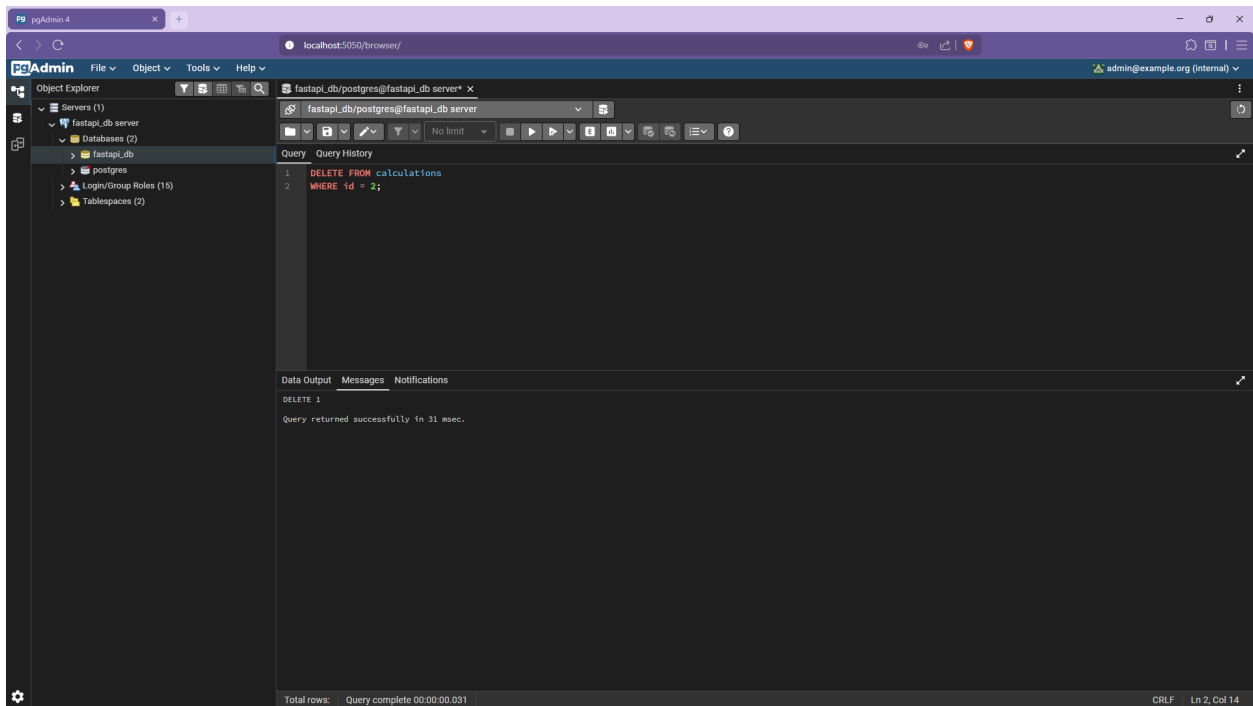


Figure 10. DELETE executed. Messages tab confirms DELETE 1.

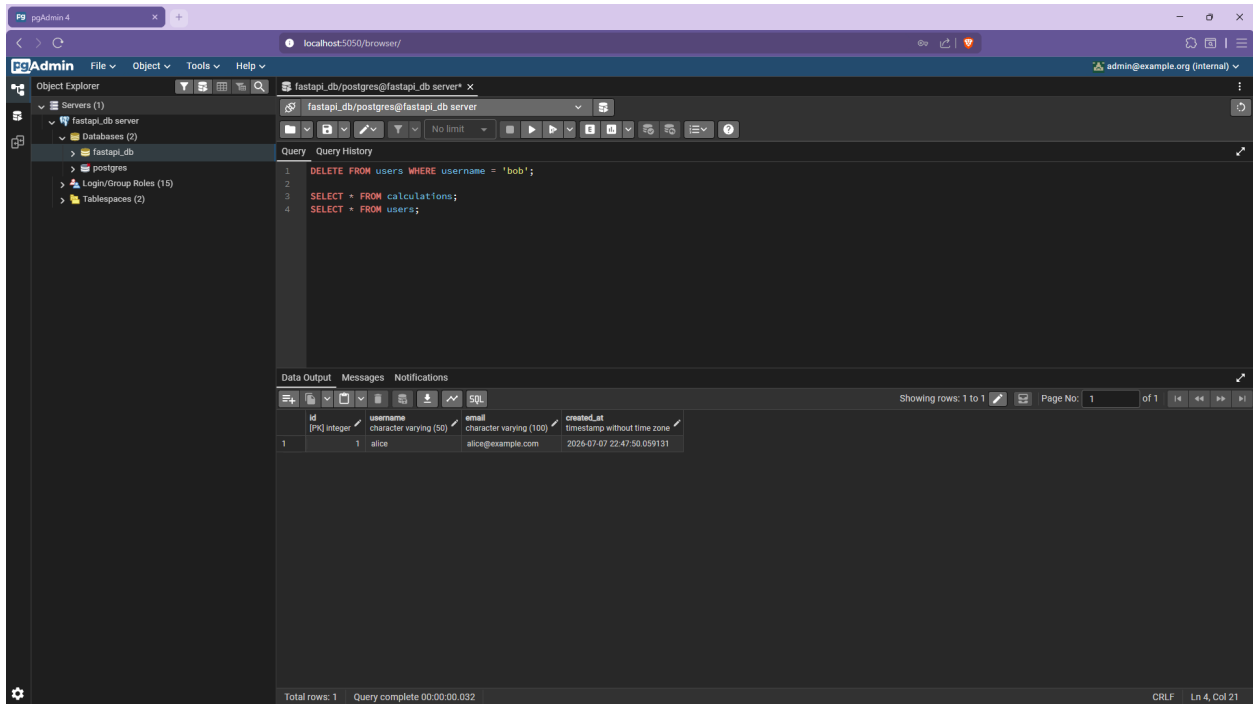


Figure 11. DELETE FROM users WHERE username = 'bob'. The users grid now contains only alice.

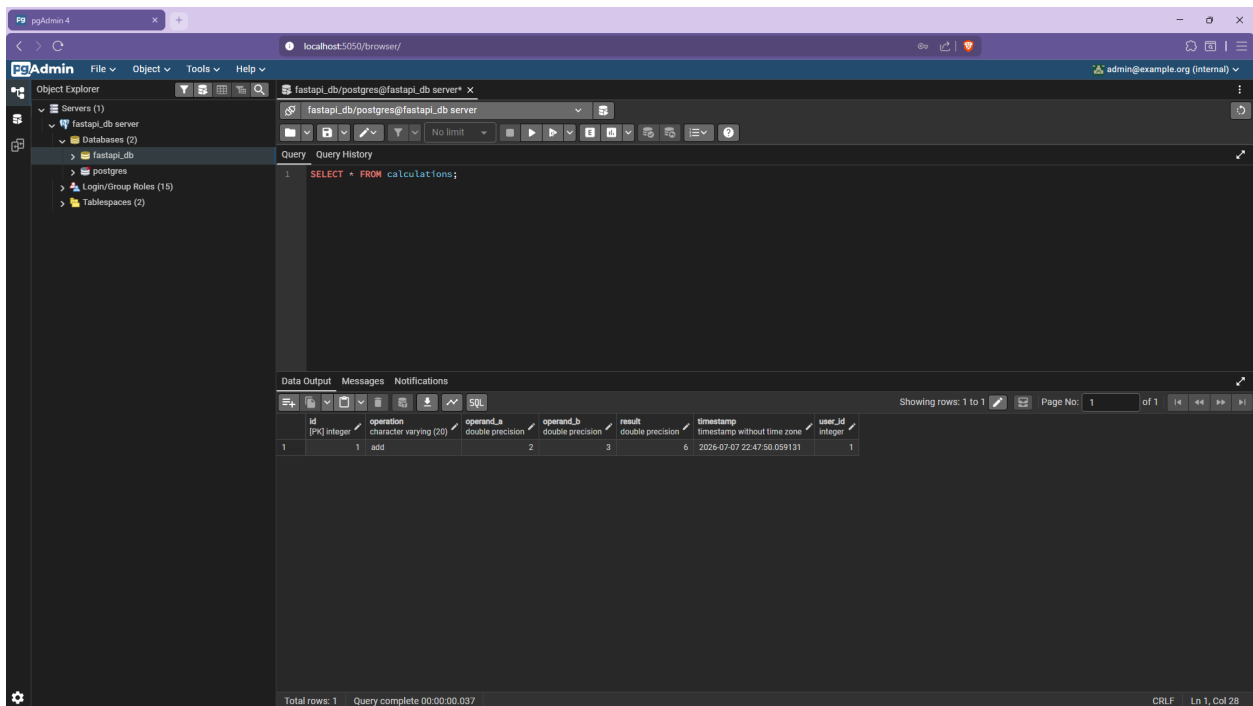


Figure 12. SELECT * FROM calculations after the cascade: a single row remains, alice's add calculation with the updated result of 6. Bob's multiply row was removed by the cascade.